

Phase 2 Define

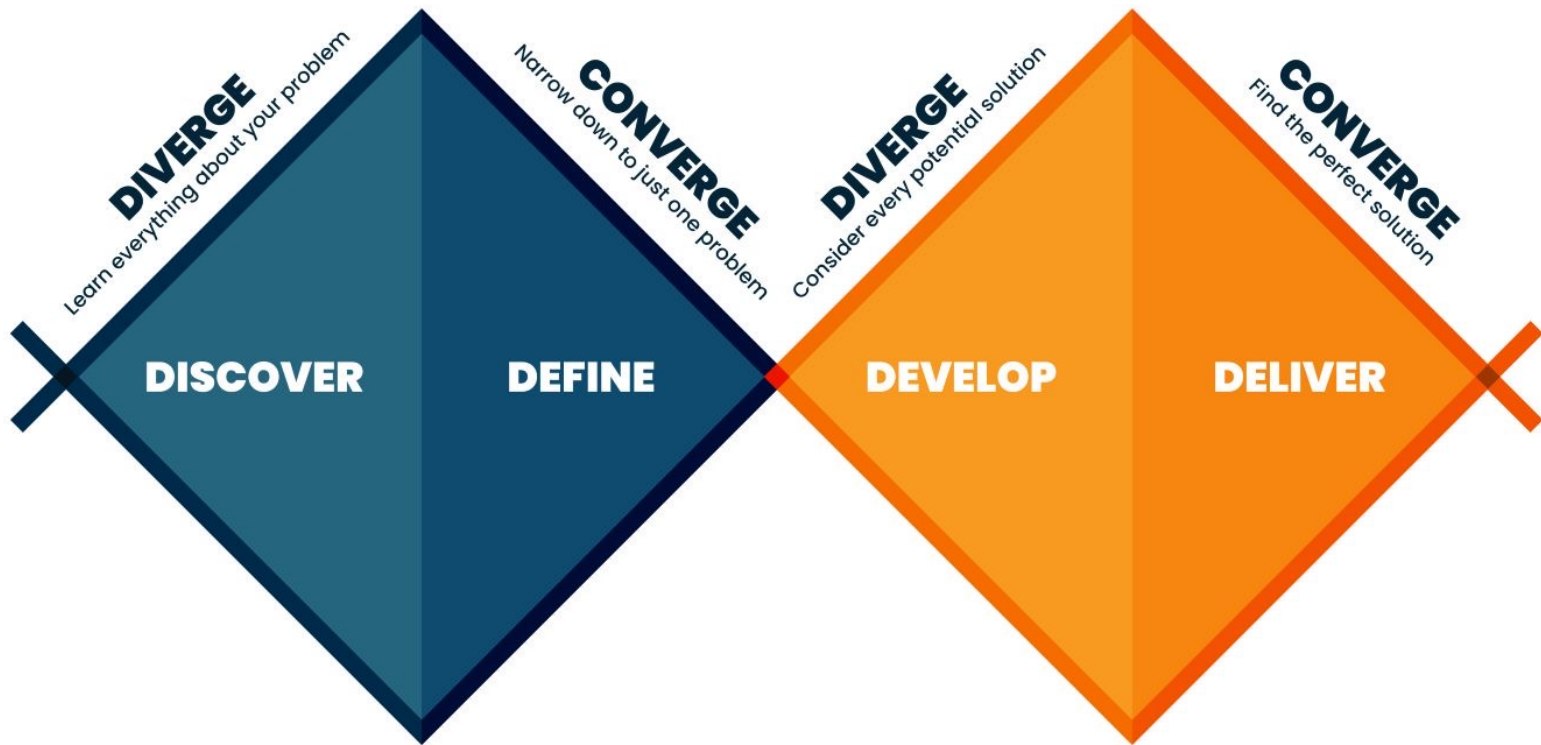
DAEN 460 Capstone Senior Design

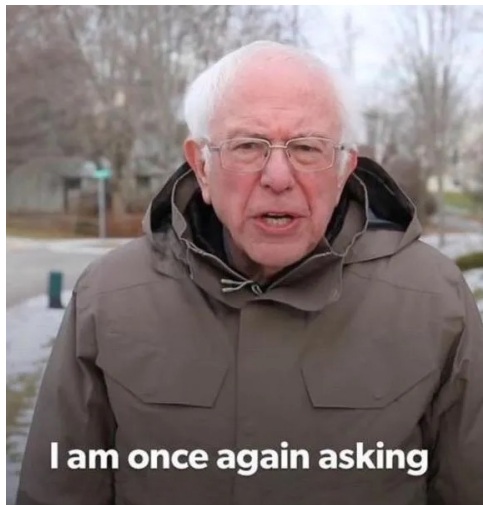
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Spring 2026



ENGINEERING
TEXAS A&M UNIVERSITY





- **Critique** your own project against **learning goals** rather than technical checklists only!
- You are **not** being graded on **reproducing** a solution but on the **quality of your thinking and decisions**!
- Peers **are not parallel workers**, but as part of a **shared system**. Shift the culture from individual performance to collective responsibility and make **collaboration** (not cooperation only)!
- **Metacognition** ...
why I'm thinking what I'm thinking → make the invisible visible!

Course Learning Outcomes *(read the syllabus!)*

Ability to:

1. **Identify, formulate, and solve** complex engineering problems by applying principles of engineering, science, and mathematics.
2. **Apply engineering design** to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.
3. **Communicate effectively** with a range of audiences.
4. **Recognize ethical and professional responsibilities** in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.
5. **Function effectively on a team** whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks and meet objectives.
6. **Develop and conduct appropriate experimentation, analyze and interpret data** and use engineering judgment to **draw conclusions**.
7. **Acquire and apply new knowledge as needed**, using appropriate learning strategies.

Today's Agenda

- **Virtual Project Showcase**
- **Communication Activity**
- **Customer Needs/Technical Objectives**
(revisions/updates)
- **Requirements/Engineering Actions**
- **Systems Modeling**
- **ABET information**
- **Phase 2 Guidelines**



Virtual Project Showcase Registration

<https://tamueps.secure-platform.com/students>

Engineering excellence. Aggie innovation. Real-world impact.

**FEBRUARY
28, 2026**

Team
Registration
Deadline



**MARCH 27,
2026**

Team Confirmation
of Attendance
Deadline



**APRIL 24,
2026**

Engineering
Project
Showcase

Required assignment for all teams, submit proof on Canvas
Project title will identify your team in published material,
ensure you get it correct!

Communication Activity

Communication is a Superpower

Communication Skills

```
graph TD; A[Communication Skills] --> B[Active Listening]; A --> C[Story Telling]; A --> D[Constructive Feedback];
```

Active Listening

I sure you've heard the saying, "We have two ears and one mouth so that we can listen twice as much as we speak." That's a key part of communication. Actually understanding who we are speaking with and then communicating our ideas once understanding theirs

Story Telling

There is a lot of talk about getting buy-in. A key part to that is telling a story. A story that moves people and helps them see how they can go from a to b.

Constructive Feedback

Giving feedback helps other people grow. But most of us are too scared to do it. We need to practice shaping messages that help people understand they can improve and we care about them

Customer Needs

(Voice of the Customer)

- By now you should have a least some customer needs identified
 - Typically, this comes from:
 - The project description
 - Discussion with the sponsor/stakeholders
- Needs should not be directed towards a potential solution
 - Often a *flaw of needs* created without a thorough technical analysis

Customer Needs

- Garvin's dimensions of quality can provide the framework to identify needs
- As you model your "system", keep these important quality dimensions in mind to identify additional needs

Product Quality	Dimension	Definition
Product based approach	Performance	The primary operating characteristics of a product.
	Features	The secondary characteristics of a product that supplement its basic functioning.
	Durability	The product's probability of failure-free performance over a specified period of time.
Manufacturing based approach	Conformance	The degree to which a product's physical and performance characteristics meet design specifications.
	Reliability	A measure of useful product life, i.e., the amount of use a customer gets from a product before it deteriorates or must be replaced.
	Serviceability	The ease, speed, courtesy, and competence of repair.
User based approach	Aesthetics	How the product looks, feels, sounds, tastes or smells, a matter of personal preferences.
	Perceived	Quality based on image, brand name, or advertising rather than product attributes and, of course, is subjectively assessed.

Requirements

(Voice of the Engineer)

- Aka Engineering Characteristics
- Requirements are objective/testable/measurable
 - *Precise, clear, concise: "The system shall..."*
- Should address one or more (typically) of the customer needs
 - *Should be developed after a good representation and analysis of the problem!*
- Demonstrate how data engineering moves from *vague stakeholder expectations* ("fast, accurate, reliable data") to **clear, testable, operational requirements tied** to infrastructure, product characteristics, and process controls

Input / Process / Output (I/P/O)

Requirements Example

- I/P/O: *There exists a need for reliable, high-quality data to support near real-time business decisions*
- **Performance**
 - What does "near real-time" mean?
 - < 1 minute latency?
 - 5 minutes?
 - Hourly refresh?
- **Quality**
 - What constitutes "high-quality data"?
 - Completeness?
 - Accuracy?
 - Consistency across systems?
 - Schema stability?

Requirements Example Cont.

- I/P/O: *There exists a need for reliable, high-quality data to support near real-time business decisions*
- **Usability**
 - Can analysts easily access the data?
 - Is it queryable without specialized engineering support?
 - Are business definitions standardized?
- **Safety / Governance**
 - Is sensitive data protected?
 - Are regulatory requirements met (PII handling, auditability)?
- **Cost**
 - Comparable to existing cloud/data warehouse budgets
 - Competitive storage and compute utilization
 - Avoid unnecessary duplication of datasets

Requirements Example Cont. ...

- **We generate a candidate set of customer needs:**
statements, which if addressed by the "*product*" would resolve the I/P/O
- Interviews and investigation; leveraging Garvin's dimensions
- Requirements are then developed from the needs; precise, clear, concise, testable, measurable
 - *Data should be available within N minutes of source system update*
 - *Data should be at least 99% complete*
 - *Critical business metrics should be consistent across dashboards*
 - *Data pipelines should not fail silently*
 - *Users should not require engineering intervention to access curated datasets*
 - *Sensitive fields should not be exposed to unauthorized users*
 - *Data refresh schedules should be transparent and visible*

- Requirements are then developed from the needs:
precise, clear, concise, testable, measurable
- Using the need as an organizing device for the requires:
Data should be available within N minutes
 - **Product**
 - End-to-end pipeline latency ≤ 5 min (source commit to warehouse availability)
 - 95th percentile latency ≤ 7 minutes
 - **Infrastructure**
 - Streaming ingestion supports $\geq 10,000$ events/sec
 - Data warehouse auto-scales to maintain query latency ≤ 3 seconds for standard BI workloads
 - **Process**
 - Monitoring alerts triggered if latency exceeds SLA (Service Level Agreement) for > 2 consecutive intervals
 - Incident response time ≤ 15 minutes

- Requirements are then developed from the needs:
precise, clear, concise, testable, measurable
- Using the need as an organizing device for the requires:
Data should be accurate and complete
 - Product
 - Record completeness $\geq 99.5\%$ per batch
 - Referential integrity violations $\leq 0.1\%$
 - Schema validation enforced at ingestion
 - Infrastructure
 - Automated data quality checks executed on every load
 - Failed quality checks block downstream publication
 - Process
 - Daily reconciliation against source system totals
 - Root cause analysis completed within 1 business day for quality breaches

- Requirements are then developed from the needs:
precise, clear, concise, testable, measurable
- Using the need as an organizing device for the requires:

Data should be secure

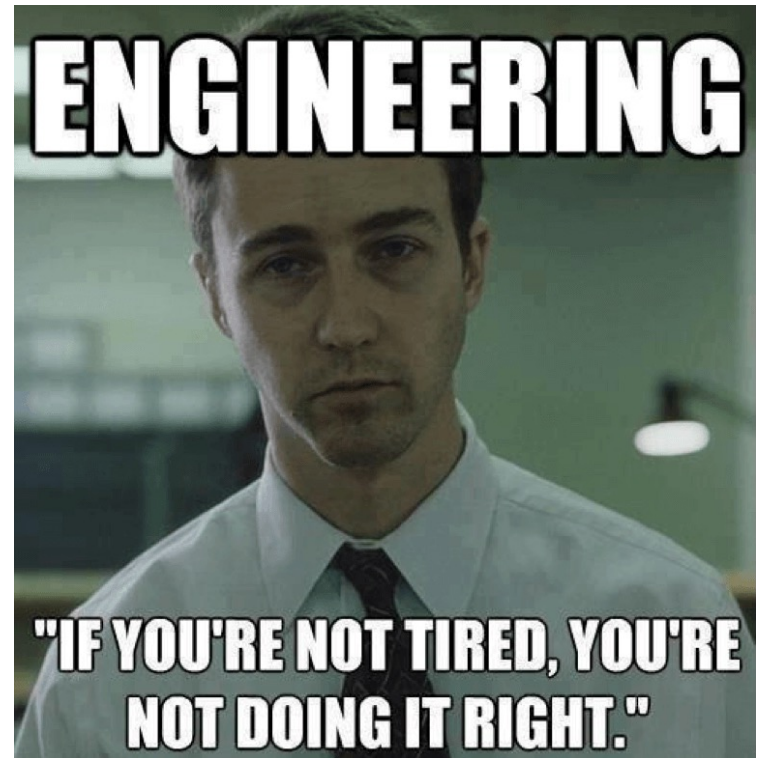
- Product
 - Role-based access control enforced at dataset and column level
 - PII fields encrypted at rest and in transit
- Infrastructure
 - Access logs retained for ≥ 365 days
 - Audit events queryable within 24 hours
- Process
 - Quarterly access review
 - Automated removal of inactive accounts after 30 days

- Requirements are then developed from the needs:
precise, clear, concise, testable, measurable
- Using the need as an organizing device for the requires:
Data platform should be cost-effective
 - Product
 - Storage cost per TB $\leq$ defined cloud budget threshold
 - Compute utilization $\geq$ 70% during peak workload windows
 - Infrastructure
 - Automatic suspension of idle clusters after 5 minutes
 - Lifecycle policies archive cold data after 180 days
 - Process
 - Monthly cost variance report delivered to stakeholders
 - Optimization review conducted quarterly

Engineering Actions

Recall "The Plan"

- This is where you start refining the plan
- What additional information is needed
- What needs to be done
 - By whom
 - By when
- What determines when the action is complete



Constraints

- Projects are subject to **constraints**
- They can be
 - Time
 - Financial
 - Resources
(personnel and/or facilities)
 - Safety
 - Other

Note: The semester should not be listed as a constraint for your project. Specific time constraints related to your sponsor might be identified, but the semester schedule creates **deadlines**, not constraints.



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Codes, Standards, and Specifications

Codes

- A code is a set of rules for performing some task, as in the local city building code or fire code.

Standards

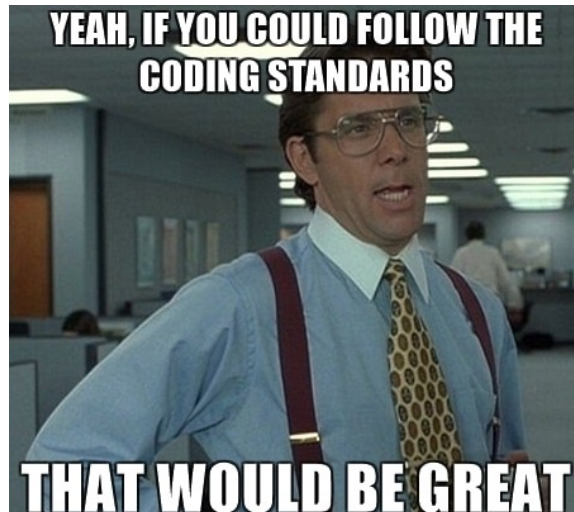
- A standard is less prescriptive and can be defined as a set of technical definitions and guidelines.

Specification

- A specification describes how a system should work, and is usually much more specific and detailed than a standard, but sometimes it is difficult to differentiate between documents that are called standards and those called specifications.

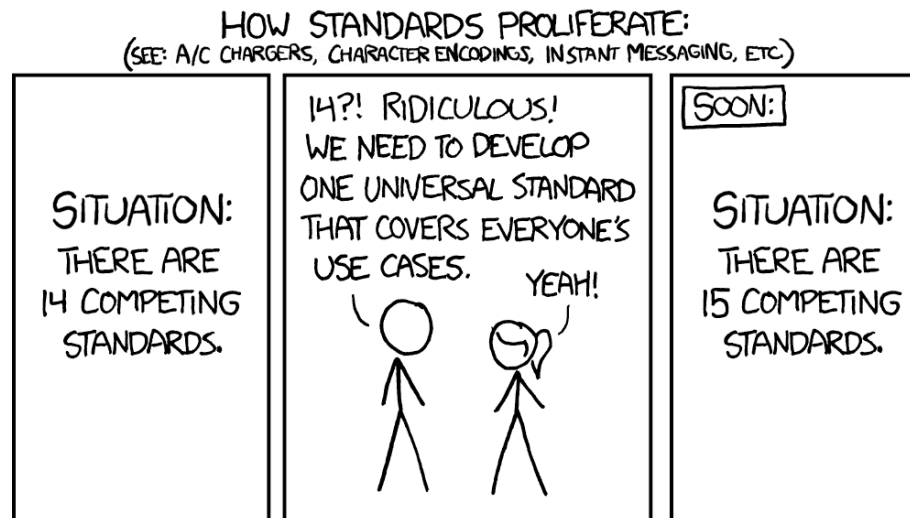
Where do you find applicable codes/standards?

- Your **project sponsor** should know specific codes/standards applicable to their systems and processes
- Your **faculty advisor** may be able to suggest codes that might apply
- **Engineering standards bodies** and organizations might have applicable codes



ANSI, NIST, ASTM, ASME, ISO

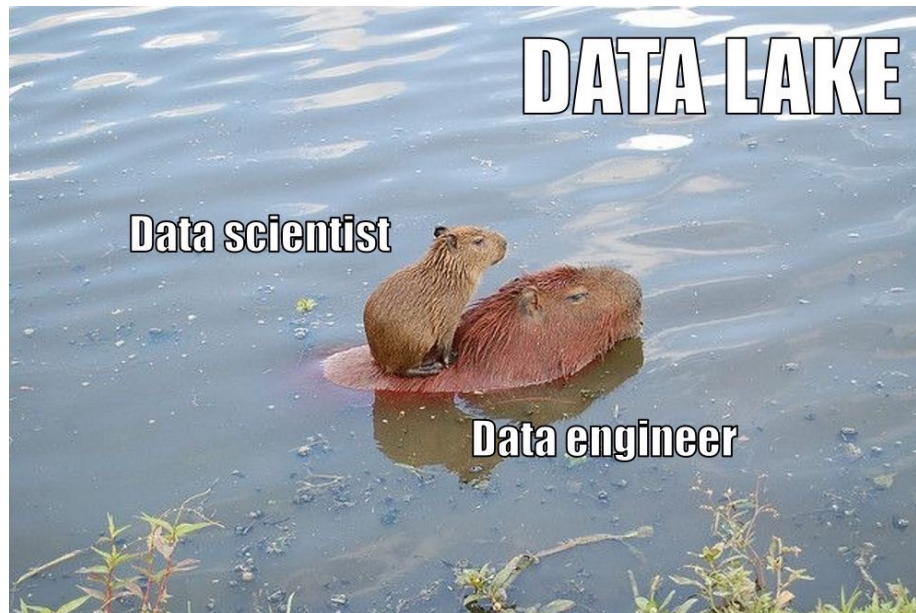
- The **American National Standards Institute** (ANSI) is the coordinating organization for the voluntary standards system for the United States (www.ansi.org)
- The standards responsibility of the U.S. government is carried out by the **National Institute for Standards and Technology** (NIST)
- **ASTM** International is the major organization that prepares standards in the field of materials and product systems
- The **ASME** prepares the well-known Boiler and Pressure Vessel Code that is incorporated into the laws of most states
- **ISO** is the international organization for standardization (www.iso.org)



Resources for Requirements and Actions

Resources are provided on Canvas to assist with requirements and actions:

- MIT Requirements Definition Presentation
- ISEN Requirements and Actions for Research Projects



Systems Modeling

- System modeling is the **process of creating abstract representations** of real-world systems to understand, analyze, design, or improve them.
- Instead of working directly with the full complexity of reality, we build **structured models** that capture the system's key components, relationships, inputs, outputs, and behaviors.
- A system can be **physical** (e.g., a manufacturing plant), **informational** (e.g., a database pipeline), **organizational** (e.g., a supply chain), or **computational** (e.g., a distributed software architecture).

Systems Modeling

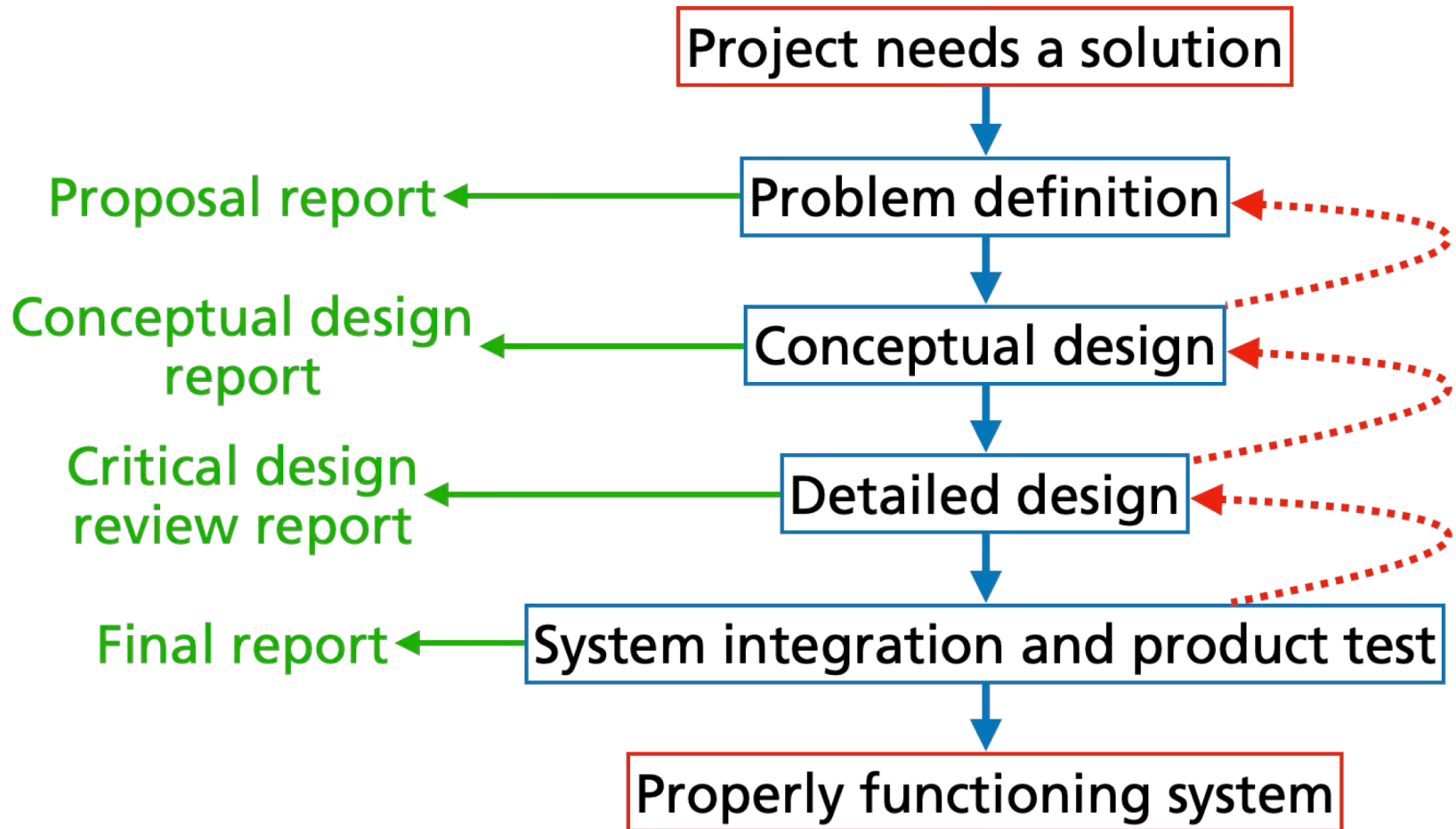
- At its core, system modeling is about **disciplined abstraction**: capturing what matters most so we can reason rigorously about complex systems!
- Common approaches include conceptual, mathematical, simulation, and data-driven models, each simplifying a system through **diagrams** (such as graphs or block diagrams), **mathematical equations** (such as differential equations), **simulations**, or **logical structures**.

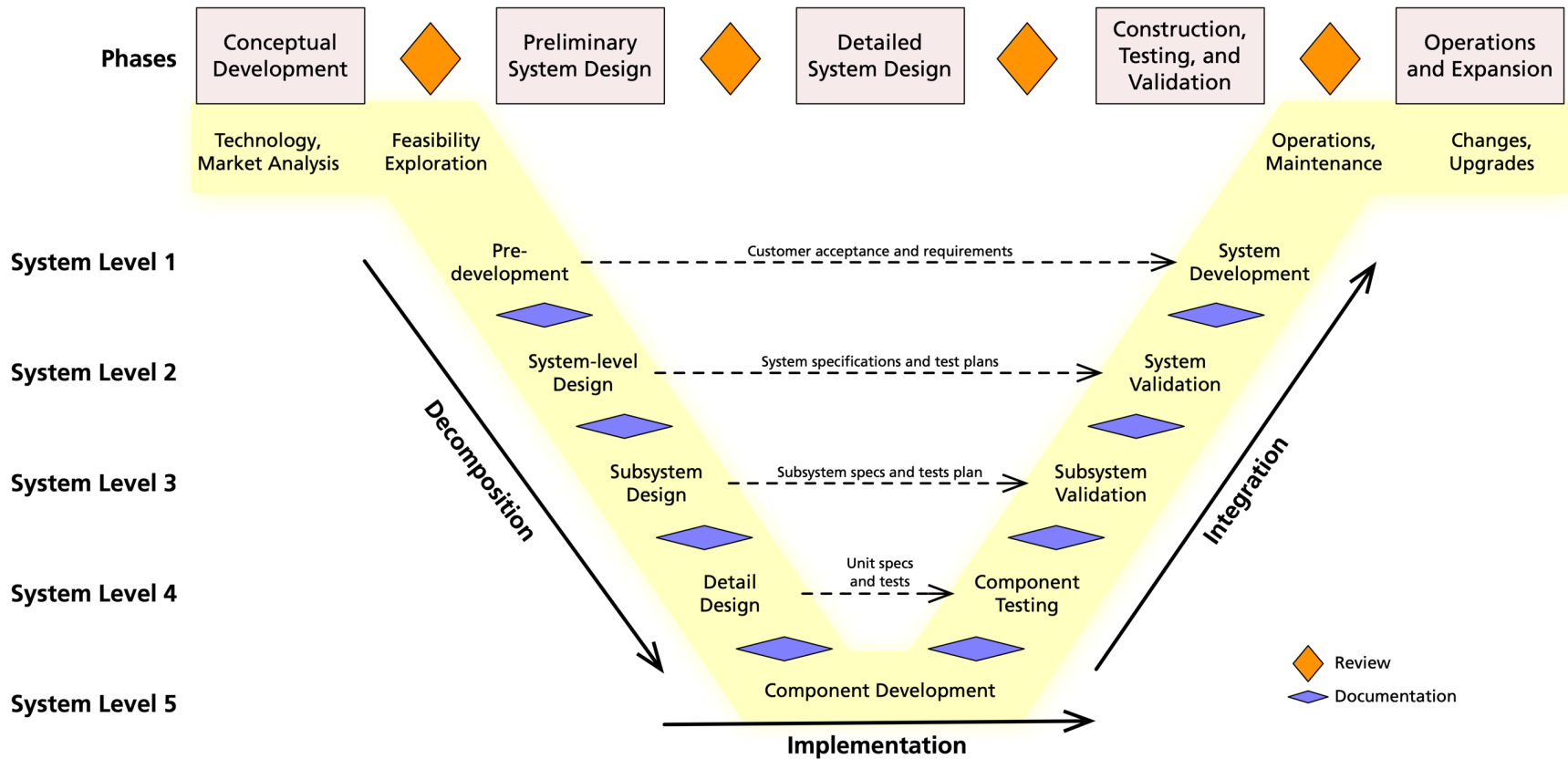
Check examples: https://en.wikipedia.org/wiki/Systems_modeling

System Modeling helps to:

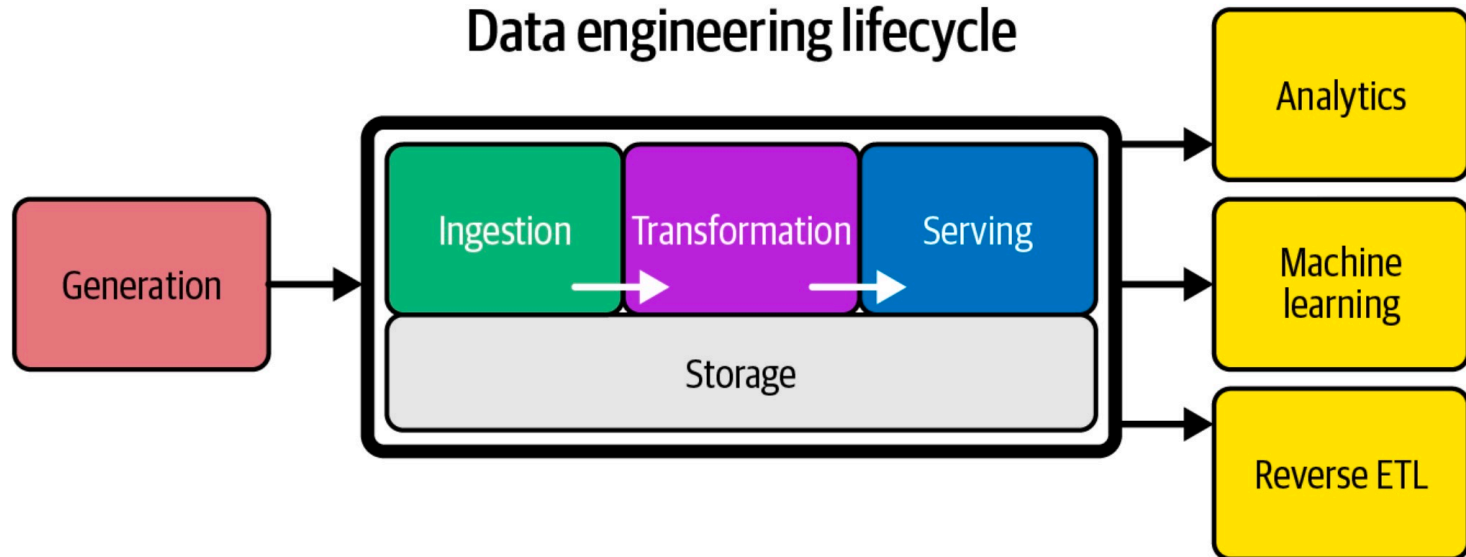
- Clarify how parts of a system interact
- Predict behavior under different conditions
- Identify bottlenecks or failure points
- Compare alternative designs before implementation
- Support optimization and decision-making

Design Process

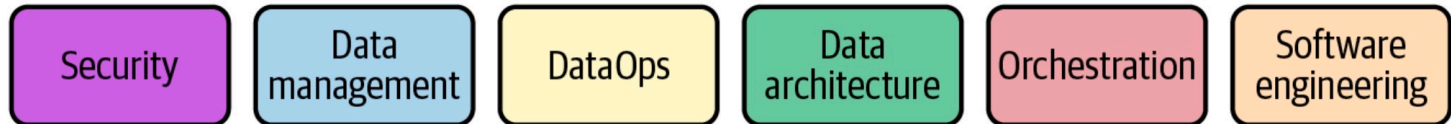




Data engineering lifecycle



Undercurrents:



ABET Accreditation and Student Outcomes

- ABET reviews the ISEN programs
- Demonstration of meeting established criteria must be documented
 - **Criterion 3, Student Outcomes**
describes the expectations for students in and graduating
 - **Criterion 5, Curriculum**
requires a culminating major engineering design experience that:
 - Incorporates appropriate **engineering standards** and **multiple constraints**
 - Is based on the **knowledge and skills** acquired in **earlier course work**
- **DAEN 460** is a primary course used for evaluation
- ABET is an important component of the value of your degree!

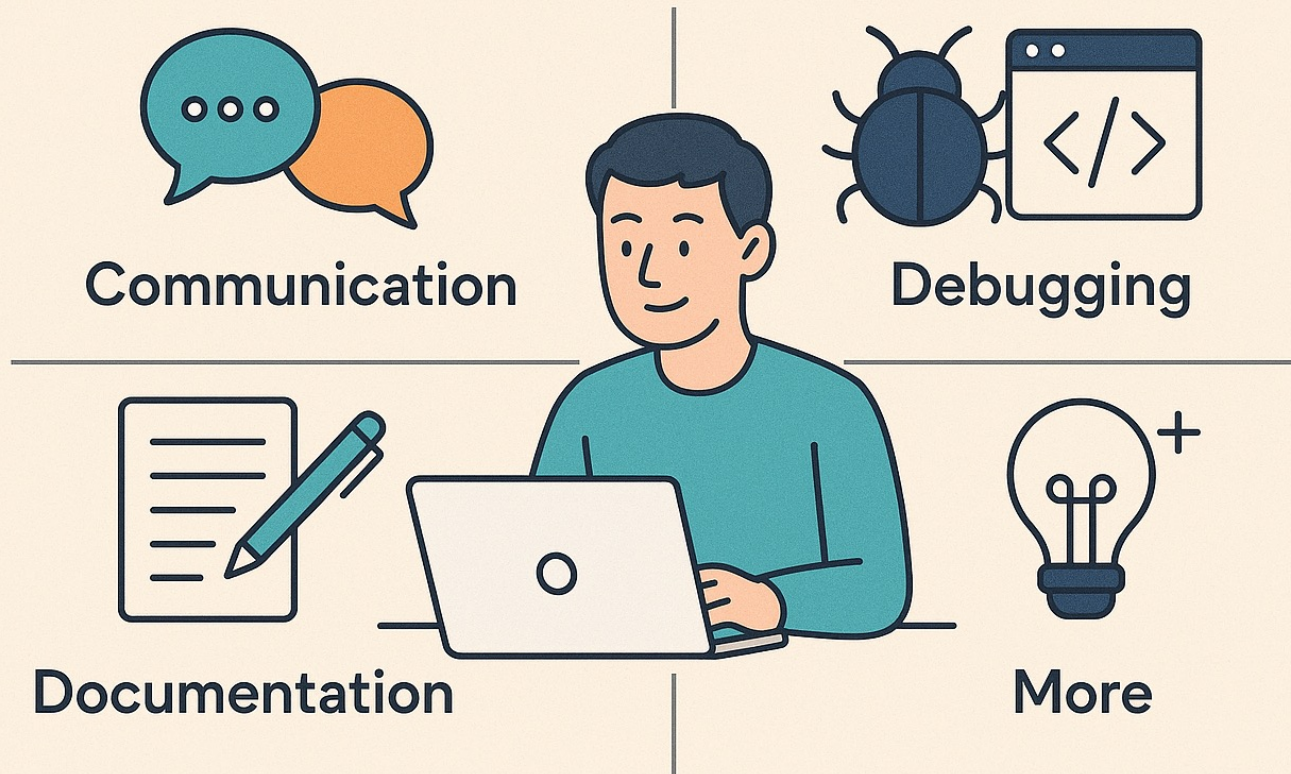
Documentation of Student Outcomes

- As part of your final report deliverables, your team will be expected to complete a questionnaire about Student Outcomes
SLIDE 3 in this presentation!
- You will reference and link to sections of the report that meet the expected outcomes
- All outcomes need to be addressed
(although some specific questions on the form may not be applicable)

Phase 2 Guidelines

On Canvas

THE HIDDEN SOFT SKILLS OF A DATA ENGINEER



Nice reading: <https://medium.com/@VijayRodrigues18/the-hidden-soft-skills-of-a-data-engineer-communication-debugging-documentation-and-more-25210d64e265>